

THE ADVERSARIAL REASONING METHOD (ARM)

A Conceptual Framework Proposal for Structured Clinical Diagnostic Reasoning

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The Problem: Diagnostic error affects an estimated 12 million U.S. outpatients annually (Singh et al., 2014), and autopsy studies show a persistent 10-40% discordance rate between clinical and post-mortem diagnoses (Hoppe et al., 2025). A literature indicates diagnostic accuracy rests primarily on retrieval of relevant knowledge rather than on reasoning strategy (Schmidt & Mamede, 2015; Norman et al., 2024). This paper treats knowledge and reasoning as interacting rather than competing. Structured reasoning support is a way to deploy existing knowledge more completely, not a substitute for acquiring it.

The Proposal: ARM is an eight-step procedural sequence (Observe, Represent, Compete, Attack, Challenge, Update, Reflect, Integrate) that is intended to walk a clinician through a structured hypothesis generation, mandatory disconfirmation, and reflection post-case as one fully connected process. Five steps are, by the paper's own account, restatements of prior work (Bordage, Croskerry, Kassirer, and Mamede & Schmidt's deliberate reflection literature). ARM's actual contribution is narrower: sequencing, a mandatory disconfirmation step, and the Reflect/Integrate stages. Each individual step targets a specific, previously documented failure mode in clinical cognition.

The Eight Steps

#	Step	Core Function	Failure Mode Addressed
1	Observe	Structured intake; separate observation from interpretation	Incomplete data gathering
2	Represent	Compress findings into a pattern-oriented summary (externalization/communication tool, not assuming an accuracy fix)	Diagnostic ambiguity
3	Compete	Generate competing hypotheses with supporting/contradicting evidence	Premature closure
4	Attack	Mandatory disconfirmation; ideally done with another clinician	Confirmation bias, anchoring
5	Challenge	Select tests by expected impact on the leading management decision, rather than breadth of hypotheses removed	Low-value testing
6	Update	Iteratively revise as new data accumulate	Diagnostic inertia
7	Reflect	Structured post-case self-evaluation	Failure to learn
8	Integrate	Long-term pattern formation across cases	Passive experience accumulation

Honest Positioning: ARM does not claim any new cognitive theory. Steps 1, 2, 3, 5, and 6 are procedural restatements of Bordage's problem representation, Kassirer's hypothetico-deductive method, Croskerry's disconfirmation strategy, and Mamede & Schmidt's deliberate reflection. The only structurally original elements are the Attack step's mandatory, fixed-position placement, and the Reflect and Integrate stages, which are not explicitly combined in this way across these bodies of work. This revision now foregrounds as ARM's primary claim to contribution.

Proposed Next Step: A small feasibility pilot (5-8 de-identified case vignettes; 2-4 raters) testing inter-rater agreement on hypothesis identification and documentation time per step. Each rater now records an initial, unstructured impression before applying ARM, enabling a within-rater comparison of whether ARM surfaces a diagnosis that rater's own snap impression would have missed. This is scoped as feasibility work, not efficacy validation.

Status: Conceptual proposal, not yet empirically tested. No claim of superiority over existing models is made. The full white paper (1.7) includes literature positioning, a worked case, pilot design, changelog, implementation constraints, and a failure log of reviewer critiques.